ASSESSMENT TOOL 1 -- PART 1

Code 1

```
# Markov Decision Process Demonstration

grid_size <- 4

start <- c(1,1)
goal  <- c(4,4)

state <- start

path <- matrix("",grid_size,grid_size)

path[start[1],start[2]] <- "S"
path[goal[1],goal[2]]   <- "G"

print("Initial Grid")
print(path)

step <- 1

cat("\nStarting Simulation\n\n")

while(!(all(state==goal))){

  row <- state[1]
  col <- state[2]

  possible <- list()

  if(row>1)
    possible$Up <- c(row-1,col)

  if(row<grid_size)
    possible$Down <- c(row+1,col)

  if(col>1)
    possible$Left <- c(row,col-1)

  if(col<grid_size)
    possible$Right <- c(row,col+1)

  best_action <- NULL
  best_distance <- 100

  for(a in names(possible)){

    next_state <- possible[[a]]

    distance <- abs(goal[1]-next_state[1]) +
      abs(goal[2]-next_state[2])

    if(distance < best_distance){

      best_distance <- distance
      best_action <- a
      state_next <- next_state

    }

  }
```

```
  reward <- -1

  if(all(state_next==goal))
    reward <- 100

  cat("Step:",step,"\n")

  cat("Current State :",state,"\n")
  cat("Action Taken  :",best_action,"\n")
  cat("Next State    :",state_next,"\n")
  cat("Reward        :",reward,"\n\n")

  state <- state_next

  step <- step+1

}
```

OUTPUT:

```
[1] "Initial Grid"
     [,1] [,2] [,3] [,4]
[1,] "S"  ""   ""   ""  
[2,] ""   ""   ""   ""  
[3,] ""   ""   ""   ""  
[4,] ""   ""   ""   "G" 

Starting Simulation

Step: 1
Current State : 1 1
Action Taken  : Down
Next State    : 2 1
Reward        : -1

Step: 2
Current State : 2 1
Action Taken  : Down
Next State    : 3 1
Reward        : -1

Step: 3
Current State : 3 1
Action Taken  : Down
Next State    : 4 1
Reward        : -1

Step: 4
Current State : 4 1
Action Taken  : Right
Next State    : 4 2
Reward        : -1

Step: 5
Current State : 4 2
Action Taken  : Right
Next State    : 4 3
Reward        : -1
```

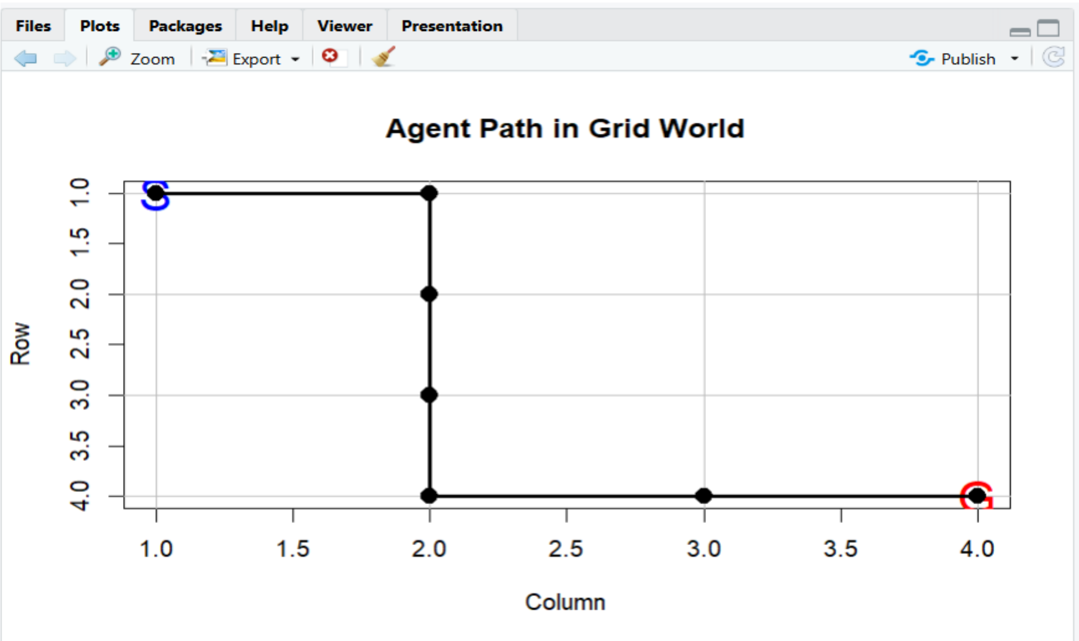
Code 2:

 # R Program to Visualize an Agent's Path in a 4×4 Grid World

```
grid_size <- 4
```

```
x <- c(1,1,2,3,4,4,4)
y <- c(1,2,2,2,2,3,4)

plot(
  1:grid_size,
  1:grid_size,
  type = "n",
  xlim = c(1,4),
  ylim = c(4,1),
  xlab = "Column",
  ylab = "Row",
  main = "Agent Path in Grid World"
)

abline(v = 1:4, col = "gray")
abline(h = 1:4, col = "gray")

text(1,1,"S",cex=2,col="blue")
text(4,4,"G",cex=2,col="red")

lines(y,x,lwd=3)
points(y,x,pch=19,cex=1.5)
```

OUTPUT:



Code 3:

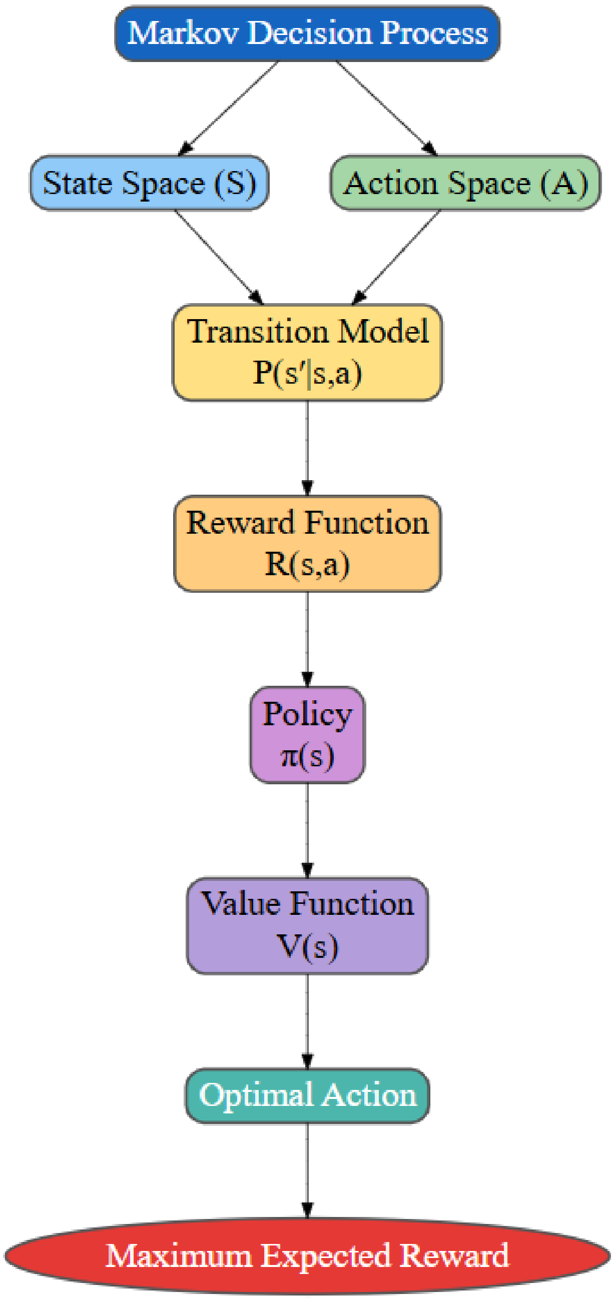
#R Program to Visualize the Markov Decision Process (MDP) Workflow Using DiagrammeR

```
library(DiagrammeR)
```

```
grViz("
digraph MDP {

graph[
layout=dot
rankdir=TB
splines=polyline
nodesep=0.6
ranksep=0.9
]

node[
shape=box
style='filled,rounded'
fontname='Times New Roman'
fontsize=24
penwidth=2
color='#555555'
]

A [label='Markov Decision Process',
fillcolor='#1565C0',
fontcolor='white']

B [label='State Space (S)',
fillcolor='#90CAF9']

C [label='Action Space (A)',
fillcolor='#A5D6A7']

D [label='Transition Model\nP(s′|s,a)',
fillcolor='#FFE082']

E [label='Reward Function\nR(s,a)',
fillcolor='#FFCC80']

F [label='Policy\nπ(s)',
fillcolor='#CE93D8']

G [label='Value Function\nV(s)',
fillcolor='#B39DDB']

H [label='Optimal Action',
fillcolor='#4DB6AC',
fontcolor='white']

I [label='Maximum Expected Reward',
shape=ellipse,
fillcolor='#E53935',
fontcolor='white']

A->B
A->C

B->D
C->D

D->E
E->F
```

```
F->G
G->H
H->I

}
")
```

OUTPUT :


Markov Decision Process
State Space (S)
Action Space (A)
Transition Model
P(s′|s,a)
Reward Function
R(s,a)
Policy
π(s)
Value Function
V(s)
Optimal Action
Maximum Expected Reward